

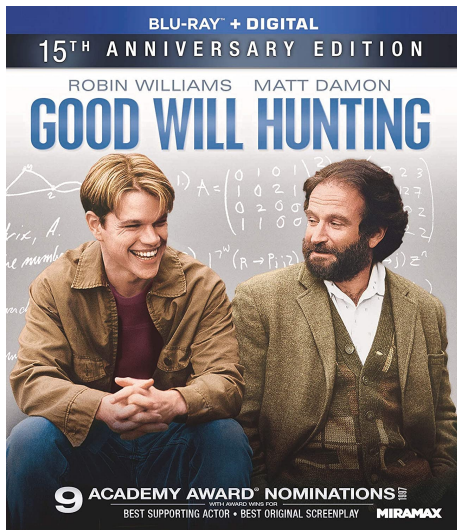
9-trees: 树 (Trees)

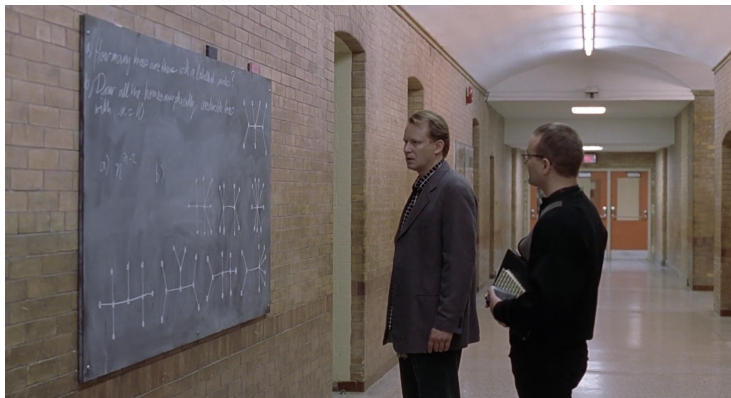
魏恒峰

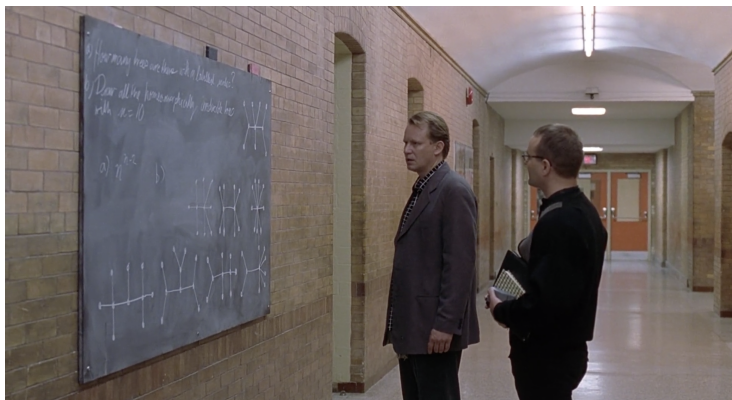
hfwei@hnu.edu.cn

2026 年 05 月 12 日

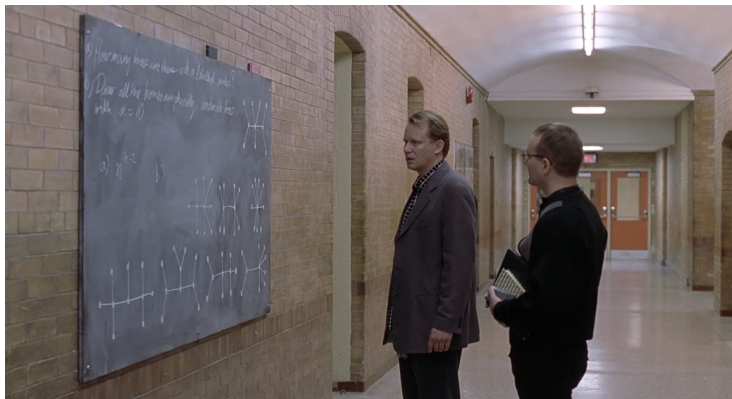




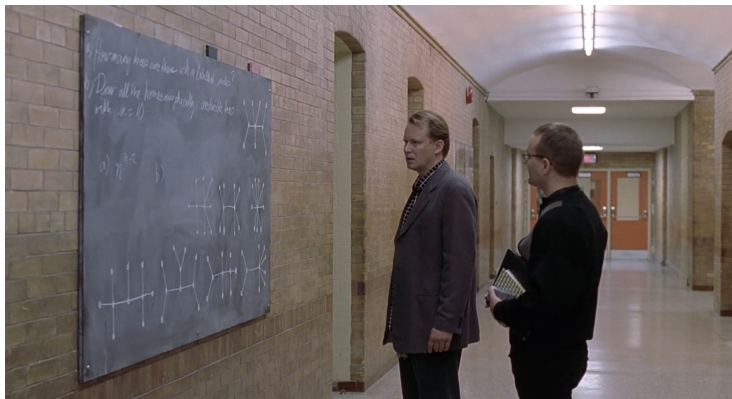




- (1) How many **trees** are there with n labeled vertices?
- (2) Draw all homomorphically **irreducible** trees with $n = 10$.



- (1) How many **trees** are there with n labeled vertices? n^{n-2}
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Definition (Tree (树))

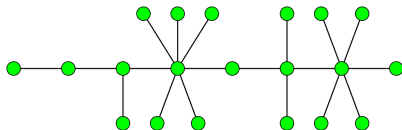
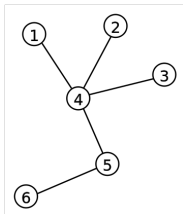
A **tree** is a **connected acyclic undirected** graph.

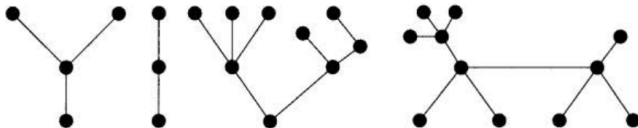
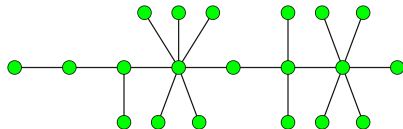
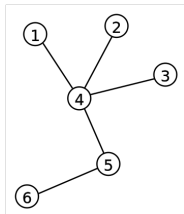
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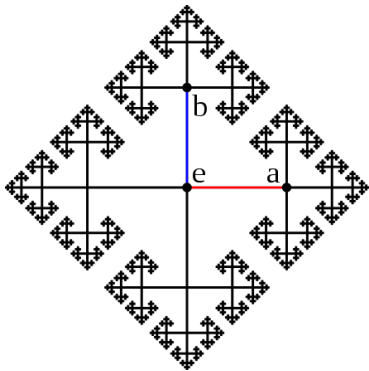
A **tree** is a **connected acyclic undirected** graph.

Definition (Forest (森林))

A **forest** is a **acyclic undirected** graph.







Cayley Graph (4-regular tree)

Definition (Internal Vertex (内部顶点); Leaf (叶子))

In a tree T with ≥ 2 vertices, for a vertex v in T , if

$$\deg(v) = 1$$

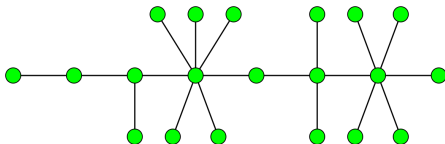
then v is called a **leaf**; otherwise, v is an **internal vertex**.

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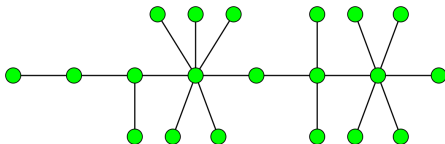


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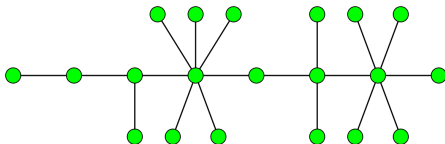
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Any tree T with ≥ 2 vertices contains ≥ 1 leaf.

Otherwise, $\forall v \in V \deg(v) \geq 2 \implies T$ has cycles.

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Consider the **two endpoints** of any **maximal** path in T .

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Consider the **two endpoints** of any **maximal** path in T .

They are leaves of T .

Lemma

*Deleting a **leaf** from a tree T with n vertices produces a **tree** with $n - 1$ vertices.*

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A leaf does *not* belong to any paths connecting two other vertices.

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A leaf does *not* belong to any paths connecting two other vertices.

This lemma can be used in induction for trees!

Theorem ((We call it) Characterization of Trees)

Let T be an undirected graph with n vertices.

Then the following statements are *equivalent*:

- (1) T is a tree;
- (2) T is acyclic, and has $m = n - 1$ edges;
- (3) T is connected, and has $m = n - 1$ edges;
- (4) T is connected, and each edge is a *bridge*;
- (5) Any two vertices of T are connected by exactly one path;
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$$(1) \implies (2) \implies (3) \implies (4) \implies (5) \implies (6) \implies (1)$$

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For $T' = T - v$, $m(T') = (n - 1) - 1 = n - 2$.

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$$m(T) = (n - 2) + 1 = n - 1.$$

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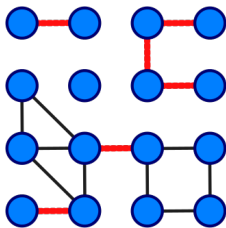
$$m(T) = \sum_{i=1}^k m(T_i) = n - k \neq n - 1.$$

Theorem (Characterization of Trees)

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Definition (Bridge (桥))

A **bridge** of a graph G is an **edge** e such that

$$c(G - e) > c(G).$$

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$$m(T - e) = (n - 1) - 1 = n - 2.$$

$T - e$ must be disconnected.

Theorem (Characterization of Trees)

- (4) T is connected, and each edge is a *bridge*;
- (5) Any two vertices of T are connected by exactly one path.

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Consider any two vertices u and v .

T is connected $\implies u$ and v are connected by ≥ 1 path.

If u and v are connected by two paths,
the edges on these two paths are not bridges.

Theorem (Characterization of Trees)

- (5) *Any two vertices of T are connected by exactly one path;*
- (6) *T is acyclic, but the addition of any edge creates exactly one cycle.*

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If T has a cycle C ,
any two vertices in C is connected by ≥ 2 paths.

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Consider the addition of edge $\{u, v\}$ to T .

It creates a cycle, consisting of $\{u, v\}$ and the path from u to v .

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Lemma

If two distinct cycles of a graph G share a common edge e , then G has a cycle that does not contain e .

Theorem (Characterization of Trees)

- (6) T is acyclic, but the addition of any edge creates exactly one cycle;
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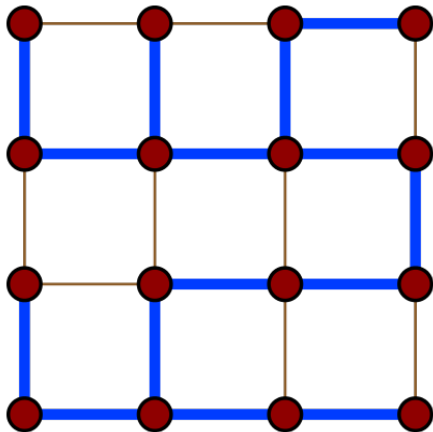
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Choose $u \in V(T_1), v \in V(T_2)$.

$T + \{u, v\}$ does **not** create cycles.

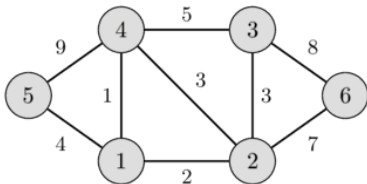


Spanning Trees (trees in graphs)

Definition (Subgraph (子图))

A graph S is a **subgraph** of G if

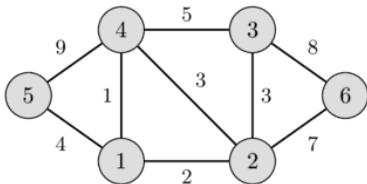
$$V(S) \subseteq V(G) \wedge E(S) \subseteq E(G) \wedge \bigcup E(S) \subseteq V(S)$$



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Definition (Induced Subgraph (诱导子图))

A graph S is an **induced subgraph** of G if S is a **subgraph** of G such that

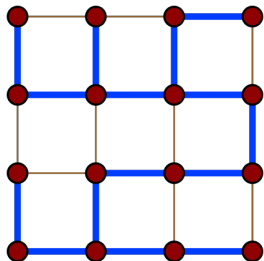
$$\{\{u, v\} \in E(G) \mid u \in V(S), v \in V(S)\} \subseteq E(S).$$

Definition (Spanning Tree (生成树))

A **spanning tree** T of an **undirected** graph G is a **subgraph** that is a **tree** with all vertices of G .

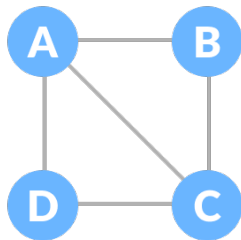
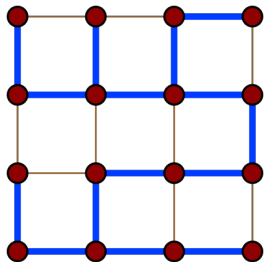
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Theorem

Every connected undirected graph G admits a spanning tree.

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Repeatedly deleting edges in cycles until the graph is acyclic.

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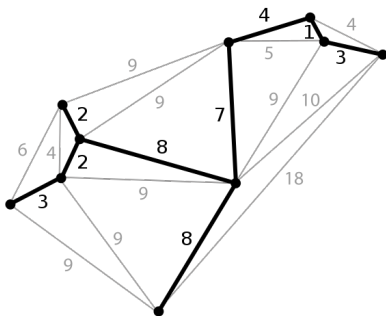
The remaining graph is a spanning tree of G .

Definition (Minimum Spanning Tree (MST; 最小生成树))

A **minimum spanning tree** T of an **edge-weighted** undirected graph G is a spanning tree with **minimum** total weight of edges.

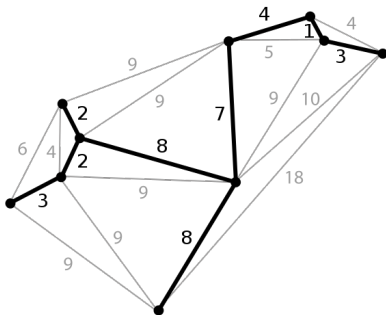
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Existence?

Uniqueness?

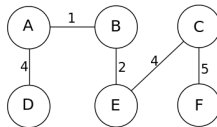
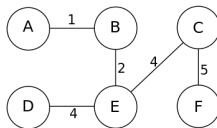
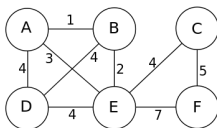
Algorithms?

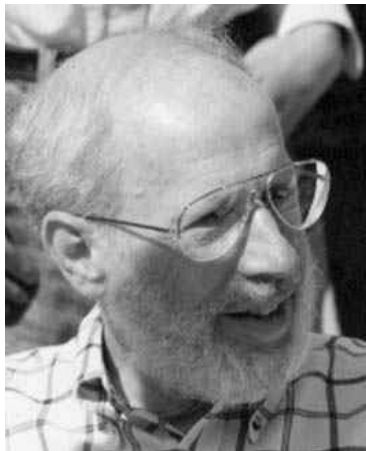
Theorem

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Theorem

Every weighted connected undirected graph G *admits* a minimum spanning tree.

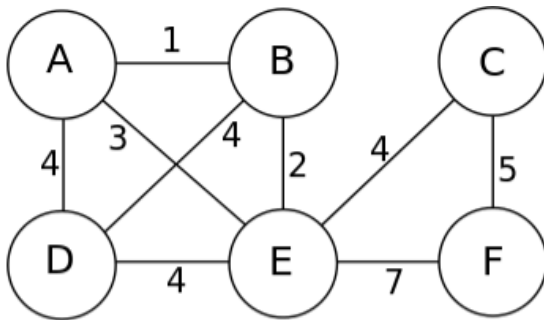




Joseph Kruskal (1928 ~ 2010)

Repeatedly adding the next lowest-weight edge that will not form a cycle until $n - 1$ edges are added.

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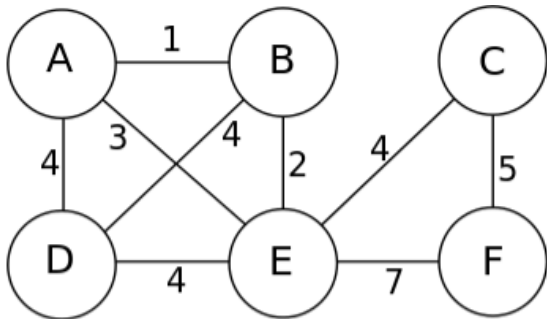




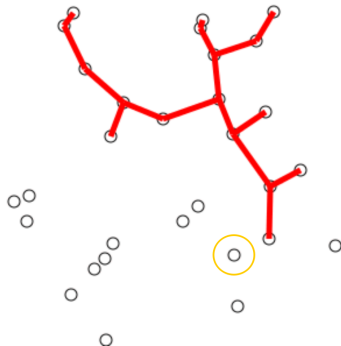
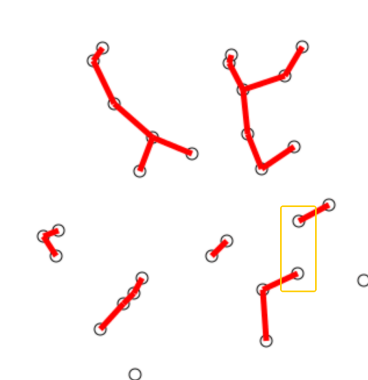
Robert C. Prim (1921 ~ 2021)

Repeatedly adding the **cheapest** possible edge
from **the partially built tree** to another vertex,
until $n - 1$ edges are added.

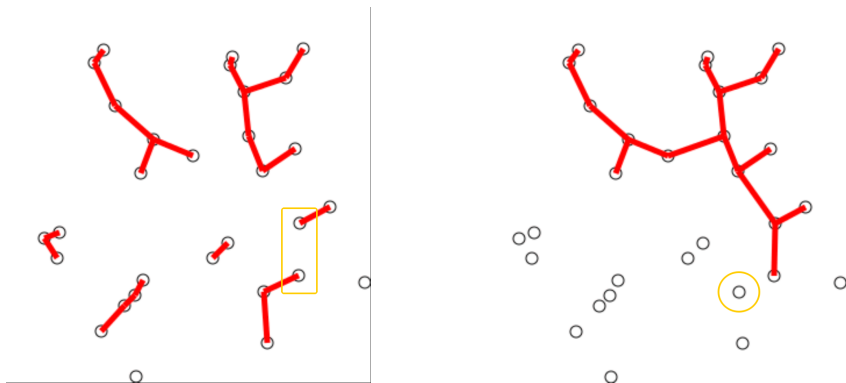
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Kruskal's and Prim's algorithms are essentially **THE SAME!**

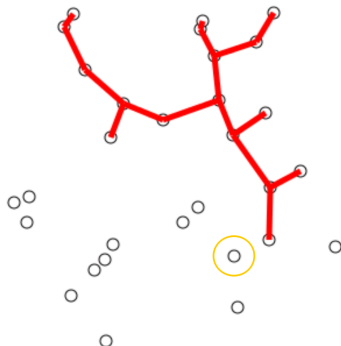
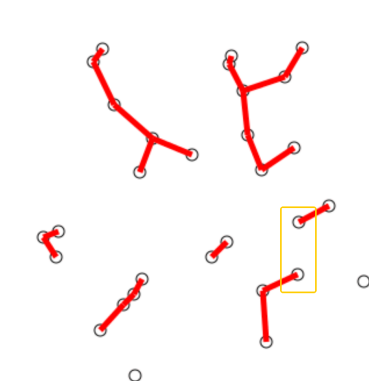


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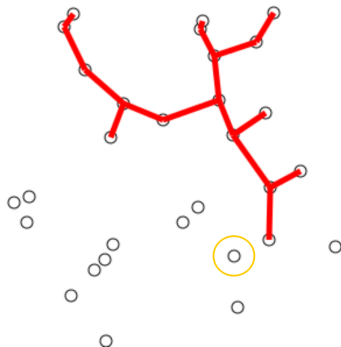
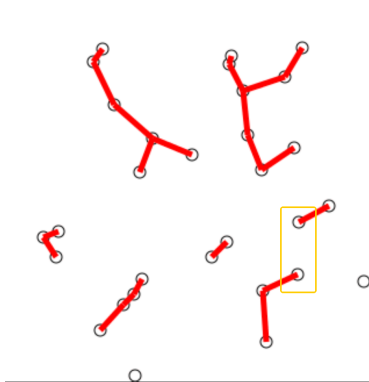


Iteratively and greedily adding “safe” edges
until $n - 1$ edges are added.

“Safe”: If X is a part of some MST T ,
then $X \cup \{e\}$ is also a part of some MST T' .



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“Safe”: the next lightest edge that does not lead to a cycle.

Theorem (Cut Property)

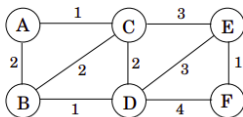
X : A part of some MST T of G

$(S, V \setminus S)$: A *cut* such that X does *not* cross $(S, V \setminus S)$

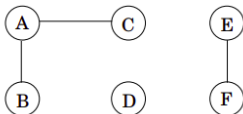
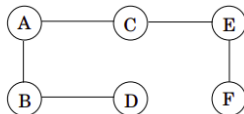
e : A *lightest* edge across $(S, V \setminus S)$

Then $X \cup \{e\}$ is a part of *some* MST T' of G .

(a)

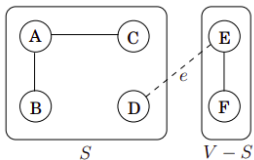
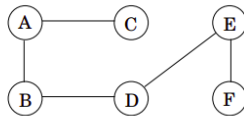


(b)

Edges X :MST T :

(c)

The cut:

MST T' :

Theorem (Cut Property)

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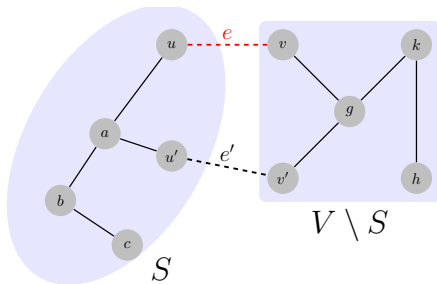
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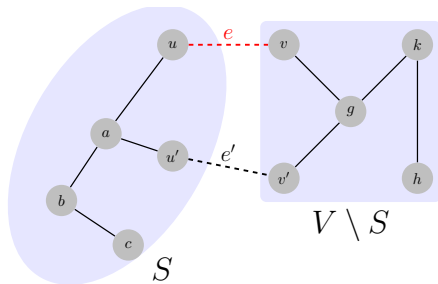
Correctness of Kruskal's and Prim's algorithms.

By Exchange Argument.

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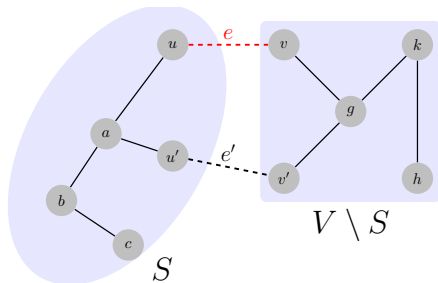


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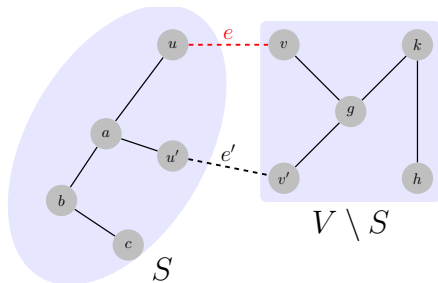
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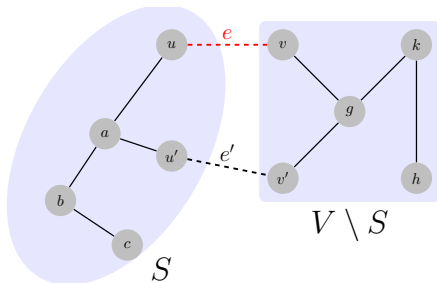
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$w(e) \leq w(e') \implies T'$ is also an MST

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“a” $\rightarrow$ “the” $\implies$ “some” $\rightarrow$ “all”

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- ▶ Let $e = (u, v)$ be *a* maximum-weight edge in C

Then $\exists$ MST T of $G : e \notin T$.

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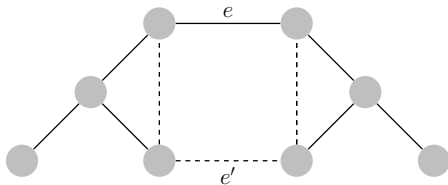
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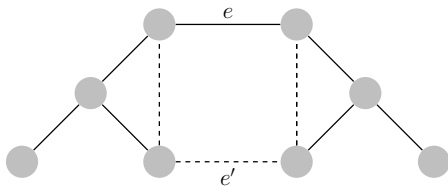
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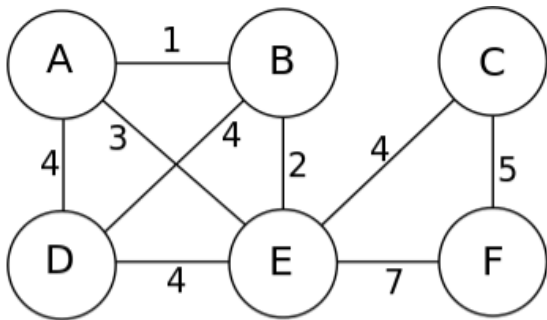
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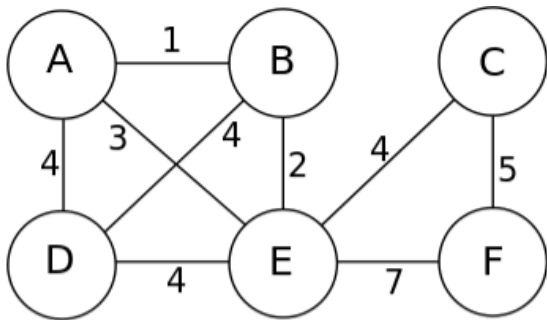
“a” $\rightarrow$ “the” $\Rightarrow$ “ $\exists$ ” $\rightarrow$ “ $\forall$ ”

Repeatedly deleting the next highest-weight edge
in some cycle until the graph is acyclic.

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Kruskal's “reverse-delete” algorithm

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Let G be an edge-weighted undirected graph.

*If each edge has a **distinct** weight, then there is a **unique** MST of G .*

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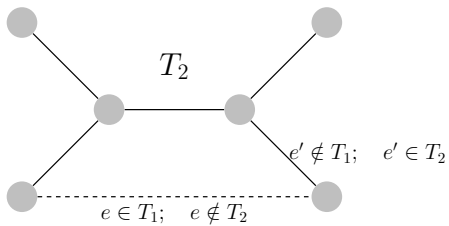
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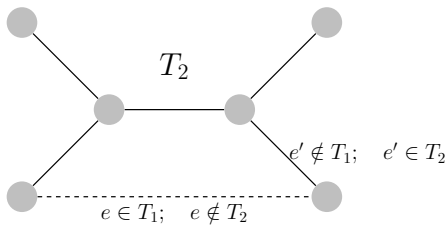
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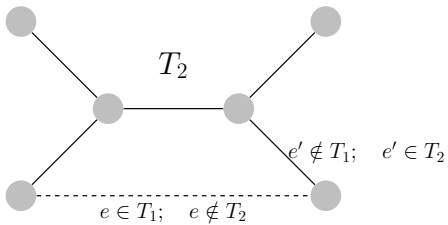
$$e = \min \Delta E$$

Suppose that $e \in T_1 \setminus T_2$ (w.l.o.g)



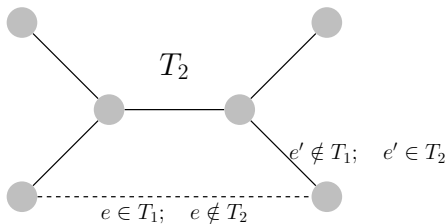


$$T_2 + \{e\} \implies C$$



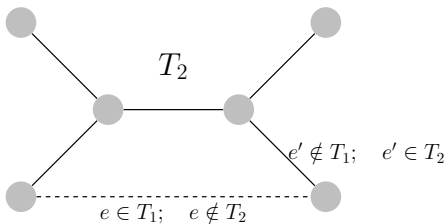
$$T_2 + \{e\} \Rightarrow C$$

$$\exists(e' \in C) \notin T_1$$



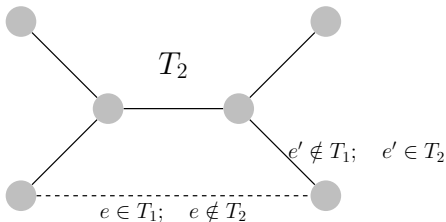
$$T_2 + \{e\} \Rightarrow C$$

$$\exists(e' \in C) \notin T_1 \Rightarrow e' \in T_2 \setminus T_1 \Rightarrow e' \in \Delta E$$



$$T_2 + \{e\} \Rightarrow C$$

$$\exists(e' \in C) \notin T_1 \Rightarrow e' \in T_2 \setminus T_1 \Rightarrow e' \in \Delta E \Rightarrow w(e') > w(e)$$



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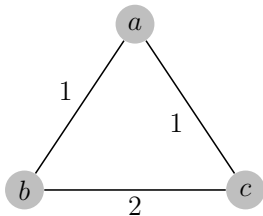
$$T' = T_2 + \{e\} - \{e'\} \Rightarrow w(T') < w(T_2)$$

Condition for Uniqueness of MST

Unique MST $\not\Rightarrow$ Distinct weights

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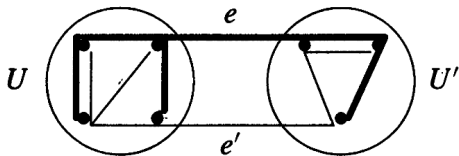


Theorem

If T, T' are spanning trees of a connected graph G and $e \in E(T) \setminus E(T')$, then there is an edge $e' \in E(T') \setminus E(T)$ such that $T - e + e'$ is a spanning tree of G .

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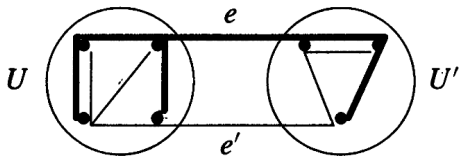
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(T is bold, T' is solid; they share two edges.)

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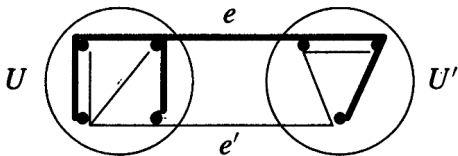


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$T - e$: two components U and U'

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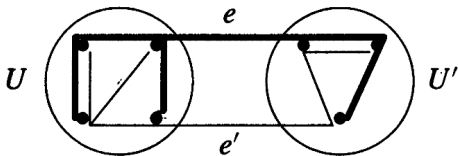
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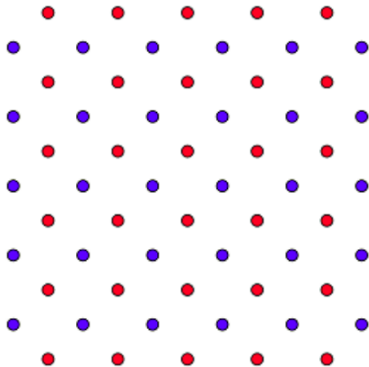
$T - e$: two components U and U'

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$T - e + e'$: a spanning tree of G

Bridg-It Game (David Gale, 1958)

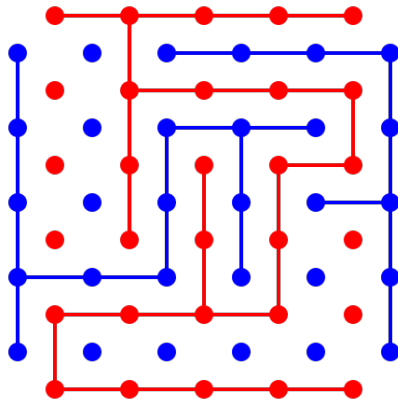
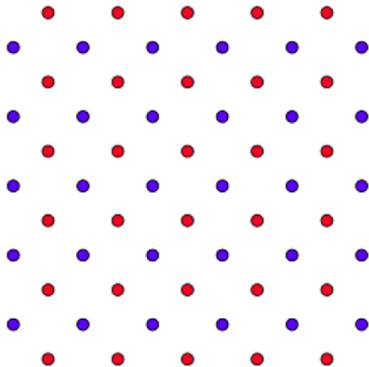
[Play with it here!](#)



5×6 vs. 6×5

Bridg-It Game (David Gale, 1958)

Play with it here!



5×6 vs. 6×5



Will Bridg-It **end in a tie**?

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No Way!

Every path from left to right **crosses** every path from top to bottom.

Does **Player 2** have a **winning strategy**?

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No! By the **strategy stealing argument**.

Does **Player 1** have a **winning strategy**?

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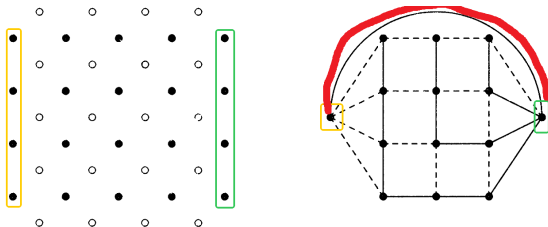
Yes!

What are **Player 1's winning strategies**?

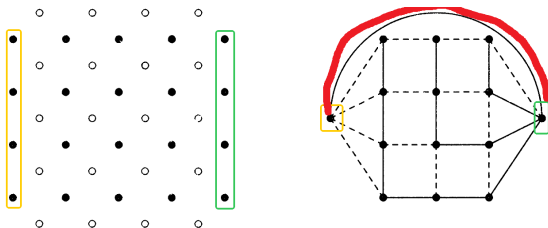
What are **Player 1's winning strategies**?

Uses **edge-disjoint spanning trees**.

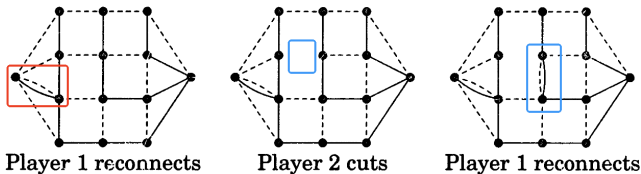
Two edge-disjoint spanning trees to connect all the dots for Player 1



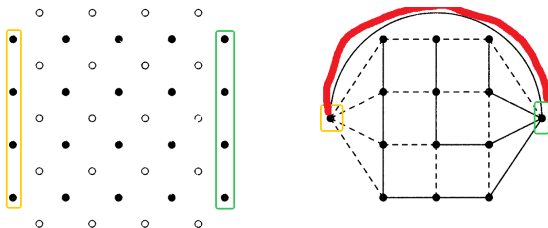
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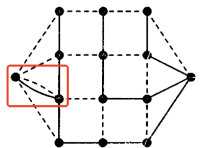
The “0-th” turn of Player 2: take away the imaginary (red) edge.



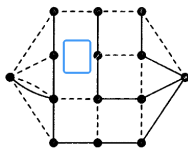
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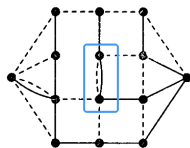
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Player 1 reconnects

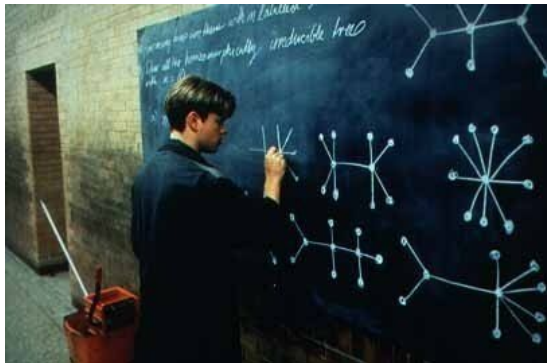


Player 2 cuts



Player 1 reconnects

Until **Player 1 has won** or when **no single edges remain to be cut**.

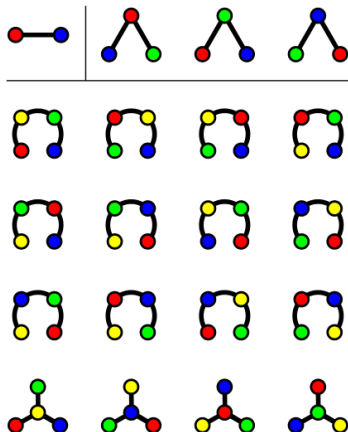


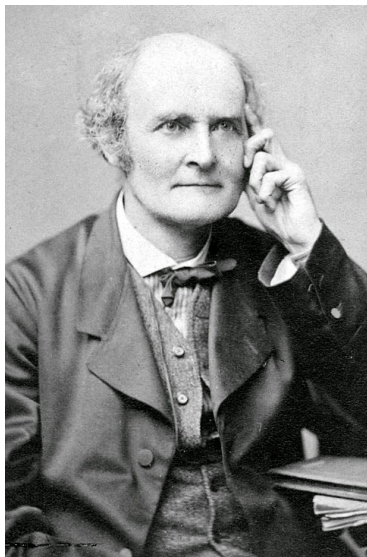
Theorem (Cayley's Formula)

The number T_n of *labeled* trees on $n \geq 2$ vertices is n^{n-2} .

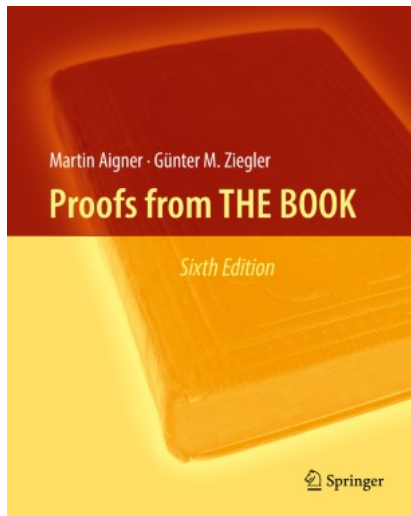
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Arthur Cayley (1821 ~ 1895)



Chapter 33: Cayley's formula for the number of trees

By Double Counting.

— Jim Pitman

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[https://en.wikipedia.org/wiki/Double_counting_\(proof_technique\)#Counting_trees](https://en.wikipedia.org/wiki/Double_counting_(proof_technique)#Counting_trees)

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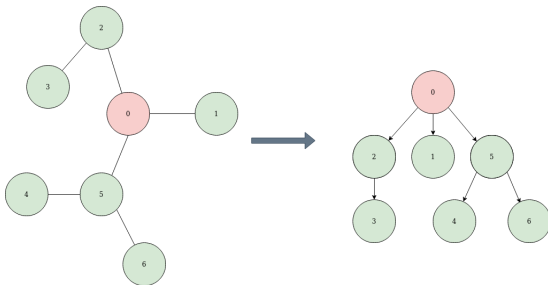
How many ways are there of forming a **directed rooted tree** from an **empty graph** by adding **directed edges** one by one?

Definition (Rooted Tree (有根树))

A **rooted tree** is a **tree** where one vertex has been **designated the root**.

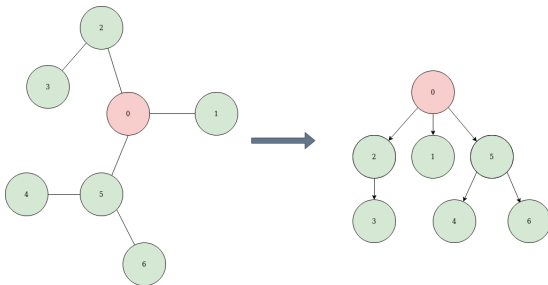
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Definition (Directed Rooted Tree (有向有根树))

A **directed rooted tree** is a **rooted tree** where all edges directed **away from** or **towards** the root.

How many ways are there of forming a **directed rooted tree** from an **empty graph** by adding **directed edges** one by one?

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Choose one of the T_n labeled trees on n vertices.

Choose one of its n vertices as root.

Choose one of the $(n-1)!$ possible sequences in which to add its $n-1$ directed edges.

$$T_n n(n-1)! = T_n n!$$

Suppose that we have added $n - k$ directed edges.

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We obtain a rooted forest with k trees.

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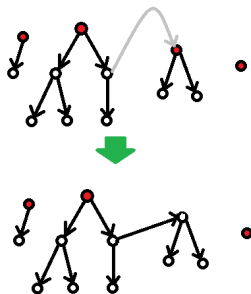
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There are $n(k - 1)$ choices for the next edge to add.

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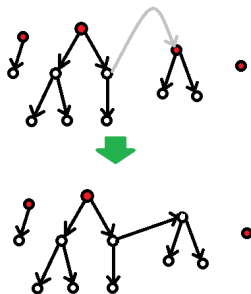
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$$\prod_{k=n}^2 n(k-1) = n^{n-1}(n-1)! = n^{n-2}n!$$

$$T_n n! = n^{n-2} n!$$

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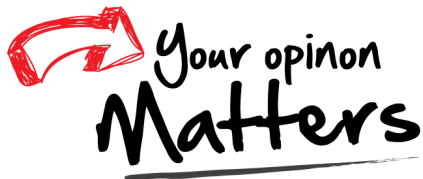
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$$T_n n! = n^{n-2} n!$$

$$T_n = n^{n-2}$$



Thank
You!



hfwei@hnu.edu.cn